

1. DEFINITION

DEFINITION
Glycerol monolinoleate EP / Glyceryl monolinoleate NF / ChP

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Glyceryl Linoleate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	CORN GLYCERIDES (UNII: 1DAF35W3S2) Or: GLYCERYL MONOLINOLEATE (UNII: 4763AXI84L)
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Alcoholysis reaction between Glycerol and refined Corn oil

COMPOSITION	
Well defined multi-constituent substance composed of mono-, di- and triglycerides of Linoleic acid (C18:2).	
Full Labeling (Key letters system) A₁ ≥ 75.0 ; 50.0 ≤ A₂ < 75.0 ; 25.0 ≤ B < 50.0 ; 10.0 ≤ C < 25.0 ; 5.0 ≤ D < 10.0 ; 1.0 ≤ E < 5.0 ; 0.1 ≤ F < 1.0 ; G < 0.1	
Glyceryl Linoleate	A ₁ ≥ 75.0%
Other constituents (i.e. incidental ingredients)	
None	



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4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	5797
Canada MF (Type III – Excipient)	/
Other countries	/
For submission of the DMF in another country, please contact your local Gattefossé representative	

CHINA BUNDLING REVIEW			
Generic name Chinese name	Registration N°	Company	CFDA/CDE Reference
Glyceryl monolinoleate 单亚油酸甘油酯	F20180000863	Gattefossé SAS	Raw material approved for use in marketed formulations in Joint-Review with drug applications.
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Glycerol Monolinoleate
USP/ NF Current edition	Glyceryl Monolinoleate
ChP Current edition	Glyceryl Monolinoleate



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6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓	✓(5) USFA 184.1505	✓(8)	✓(8) 40CFR180.910
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Glyceryl Linoleate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines	Topical Use Only in export only, Over the Counter, Prescription Medicines. Only for use in topical medicines for dermal application in Listed Medicines.
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Glyceryl linoleate	Roles non-medicinal: Purposes: Emulsifying Agent , Skin-Conditioning Agent - Emollient Route Of Administration: Topical
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		



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8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute Toxicity	Oral	Rat	Maisine 35-1 In suspension in pure olive oil	LD ₅₀ ≥ 20 000 mg/kg/day	Gattefosse data GAT-84114, 1984

CIR COMPENDIUM (Cosmetic Ingredient Review)

(<http://www.cir-safety.org>)

Denomination	Reference	Maximum “as used” concentration	Conclusions
Glyceryl monoesters (Glyceryl linoleate)	January 15, 2016	Glyceryl linoleate Up to 1%	Based on the available animal and clinical data included in the report, the CIR Expert Panel concludes that the following Glyceryl Monoesters (Glyceryl linoleate ...) are safe as cosmetic ingredients in the present practices of use and concentrations.

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION

INCI NAME	CAS #	EC #	Chemical name
Glyceryl Linoleate	68424-61-3 (or 26545-74-4) (or 91744-20-6) (or 85536-08-9)	270-312-1 (or 247-788-4) (or 294-582-5) (or 287-489-6)	Glycerides, C16-18 and C18-unsatd. mono- and di- (or (9Z,12Z)-octadeca-9,12-dienoic acid, monoester with glycerol) (or Glycerides, C16-18 and C18-unsatd. mono-, di- and tri-) (or Glycerides, corn-oil mono- and di-)

EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list.

EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5.

Substances without EC numbers were assigned “list numbers” in order to facilitate REACH submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.

NATIONAL INVENTORIES

INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDL)	Australia (AIIC)	China (IECSC)	Other inventories
Glyceryl Linoleate	Listed	DSL listed	Listed	Listed	Glycerides, C16-18 and C18-unsatd. mono- and di-: DSL, EINECS, NZIoC, PICCS, REACH, TCSI, TSCA, IECSC, AIIC, AREC, ECL / (Mono,di)glycerides(C14-18, C18-Unsaturated): TDCA, VNECI (or) 9,12-Octadecadienoic acid (9Z,12Z)-, monoester with 1,2,3-propanetriol: DSL, TCSI / (9Z,12Z)- octadeca-9,12-dienoic acid, monoester with glycerol: EINECS, REACH, VNECI (or) Glycerides, C16-18 and C18-unsatd. mono-, di- and tri- IECSC, DSL, NZIoC, TCSI, EINECS, REACH,



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NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
					TDCA, VNEC, AREC, ECL / Glycerides en C16-18 et insatures en C18, mono-, di- et tri-: DSL, EINECS (or) Glycerides, corn-oil mono-and di-: IECSC, AREC, ECL, EINECS, REACH, TCSI, VNECI
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting. Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation. Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Glyceryl Linoleate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

10. HISTORY OF USE

10.1. Pharmaceutical applications

Refer to the [HISTORY OF USE](#) document.

- [FDA IID references](#) exactly match the definition of the excipient MAISINE CC: GLYCERYL MONOLINOLEATE and CORN GLYCERIDES
- FDA IIG references to the most similar chemical (read-across approach) are GLYCERYL OLEATE and GLYCERYL MONOOLEATE

10.1.1. Other references

COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS		
Denomination	Category (concentration in %)	regulatory status
Glycerides of fatty acids	Suppository bases	1, 3–14, 16–26
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopeias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portuguese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian Garg et al., <i>Pharmaceutical Technology, Drug Delivery</i>, 2001, p14–24, Compendium Of Pharmaceutical Excipients For Vaginal Formulations		



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HANDBOOK OF PHARMACEUTICAL EXCIPIENT			
(Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfarb)			
DENOMINATION	FUNCTIONAL CATEGORY	REGULATORY STATUS	SAFETY
Glyceryl monolinoleate	Oleaginous vehicle Solubilizing agent	Glyceryl monolinoleate is included on nonparenteral medicines licensed in the UK and Europe. Included in the EAFUS list compiled by the FDA (mono- and diglycerides). Accepted for use as a food additive in Europe (E 471, mono- and diglycerides of fatty acids). Included in the Canadian Natural Health Products Ingredients database.	The Cosmetic ingredient Review expert Panel has assessed glyceryl Monolinoleate and concluded that the material is safe as used in cosmetics and personal care products. The European CHemicals Agency have applied no hazard classification or labels to glyceryl monolinoleate.

10.2. Food/Nutraceutical/Dietary supplement applications

Mono- and Diglycerides have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS471 Mono- and Diglycerides
E.U.	EFA (European Food additives)	E471 Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Mono- and Diglycerides
U.S.A.	GRAS US Food additives according to 21 CFR	21CFR §184.1505 Mono- and Diglycerides
JAPAN	JSFA (Japanese Standards of Food Additives)	Glycerol esters of fatty acids

HANDBOOK of FOOD ADDITIVES		
Denomination	Use	Safety
Glyceryl monolinoleate	Food additive	EP compliance; FDA 21CFR §184.1505 GRAS, FCC, E471, JSFA compliance; DMF no. 5797

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.



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